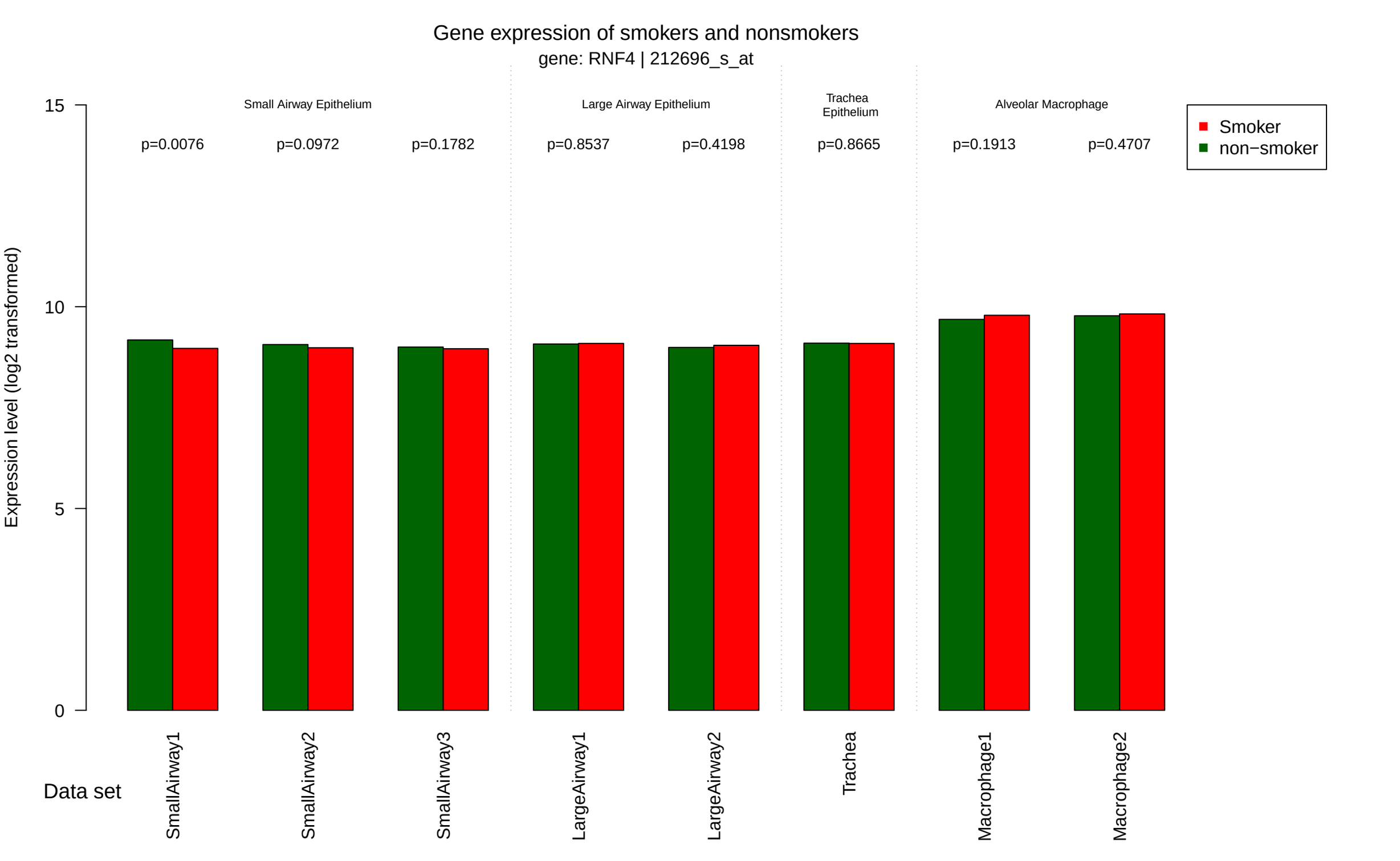
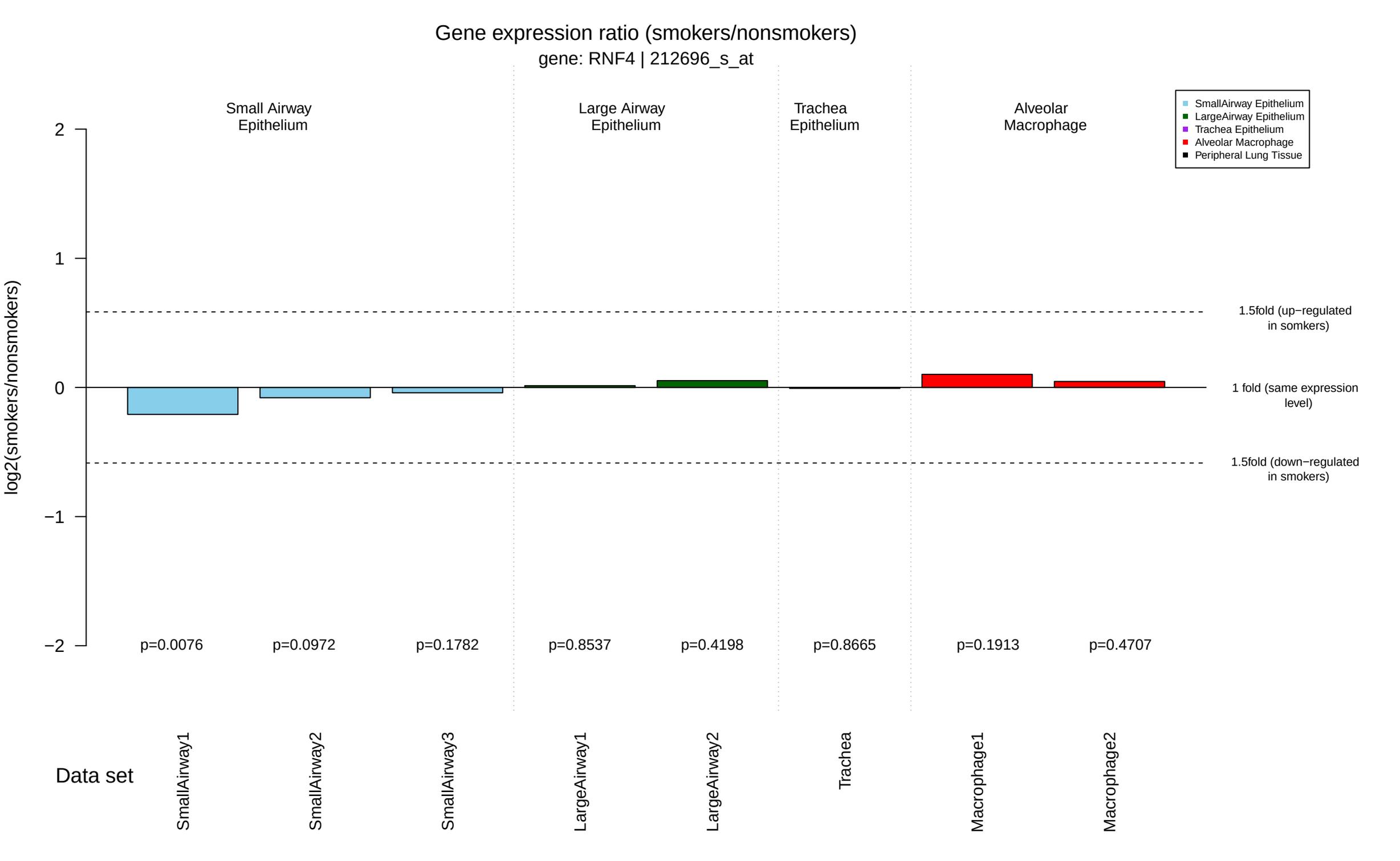
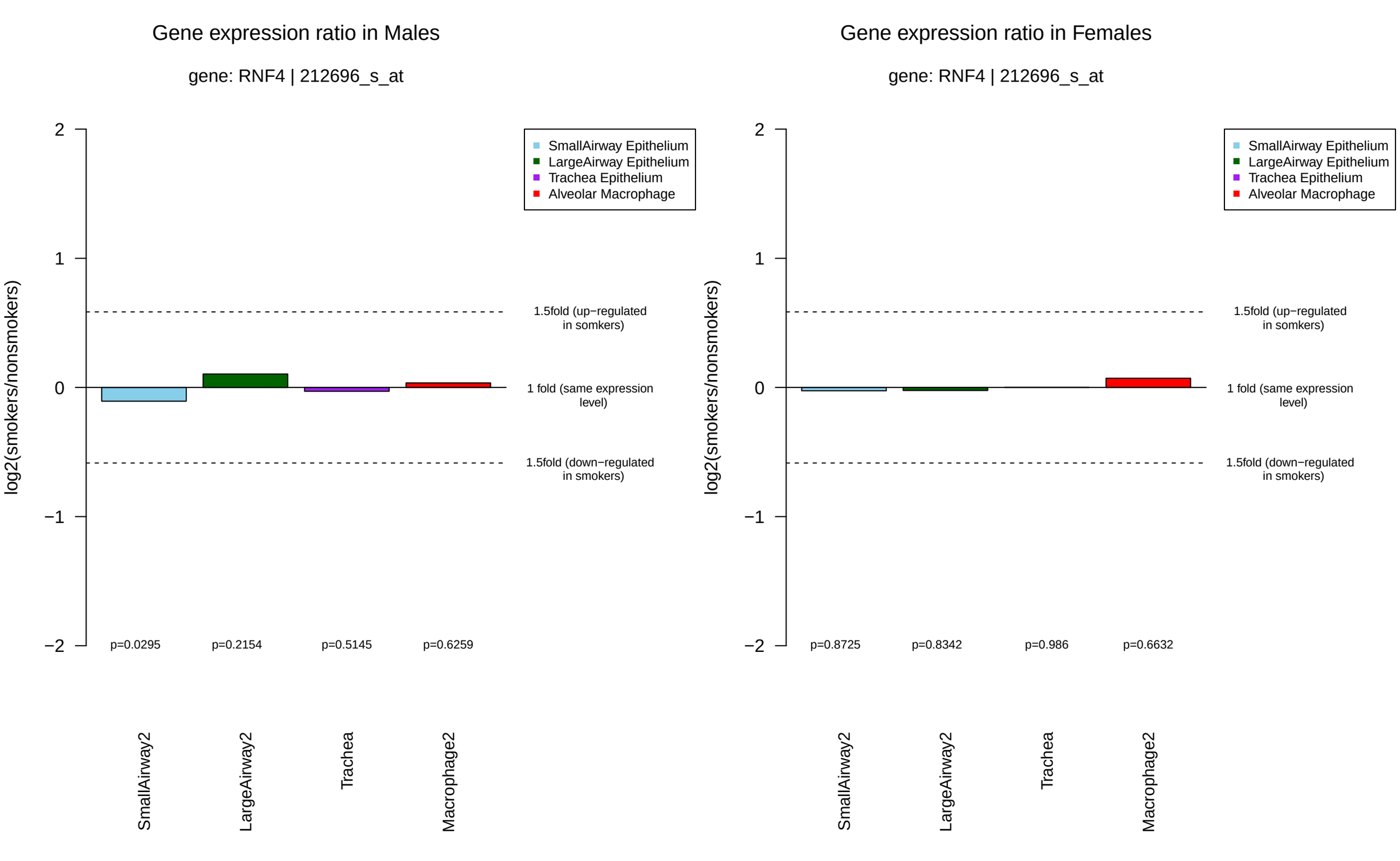
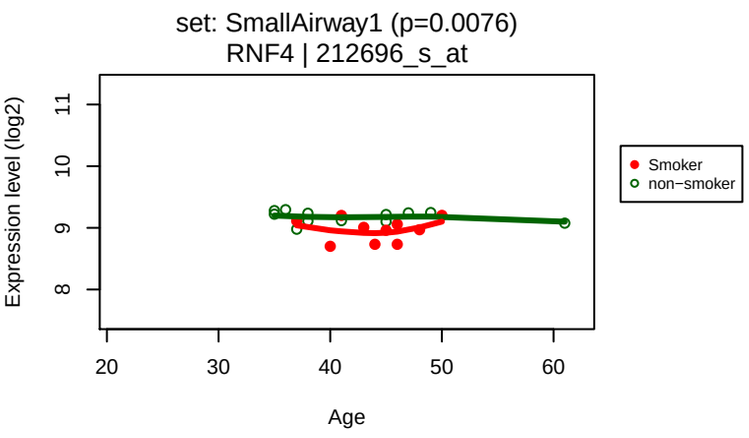
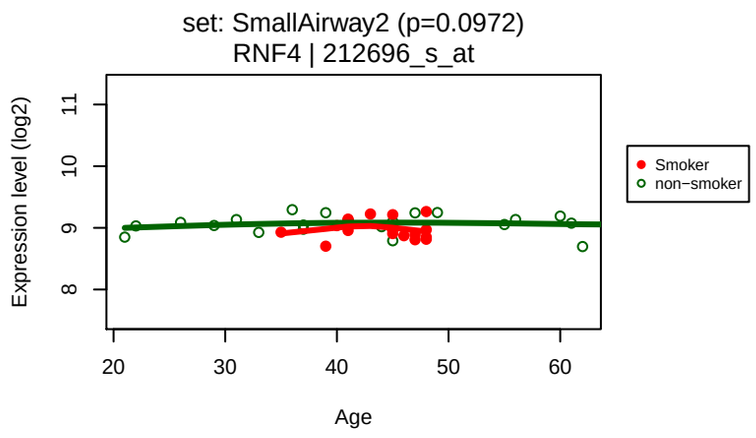
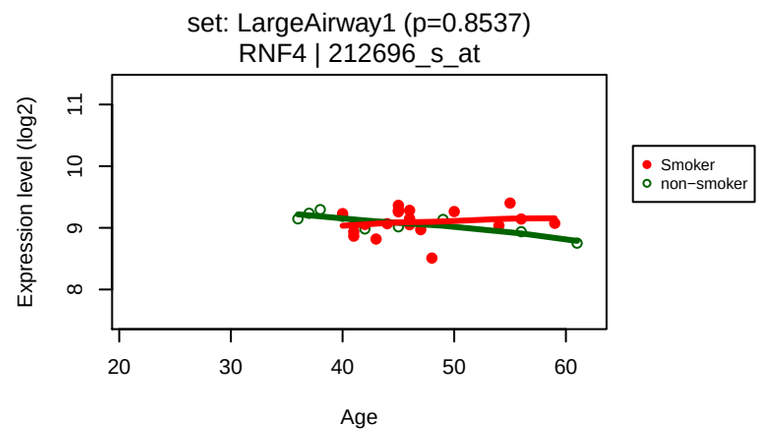

Gene expression of smokers and nonsmokers
gene: RNF4 | 212696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0076
p=0.0972
p=0.1782
p=0.8537
p=0.4198
p=0.8665
p=0.1913
p=0.4707
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

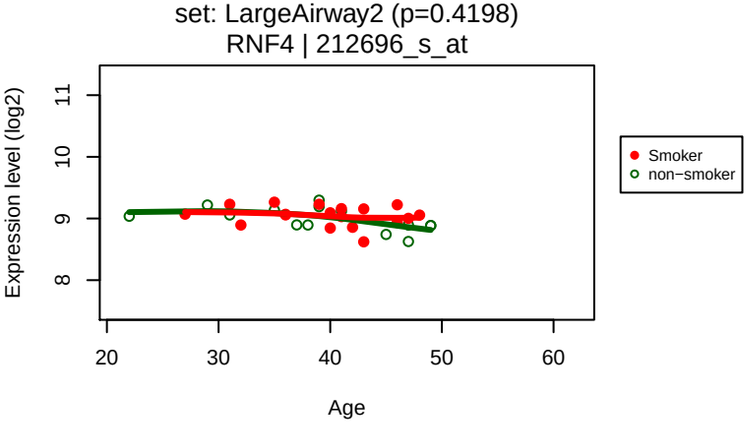

Gene expression ratio (smokers/nonsmokers)
gene: RNF4 | 212696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0076
p=0.0972
p=0.1782
p=0.8537
p=0.4198
p=0.8665
p=0.1913
p=0.4707
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

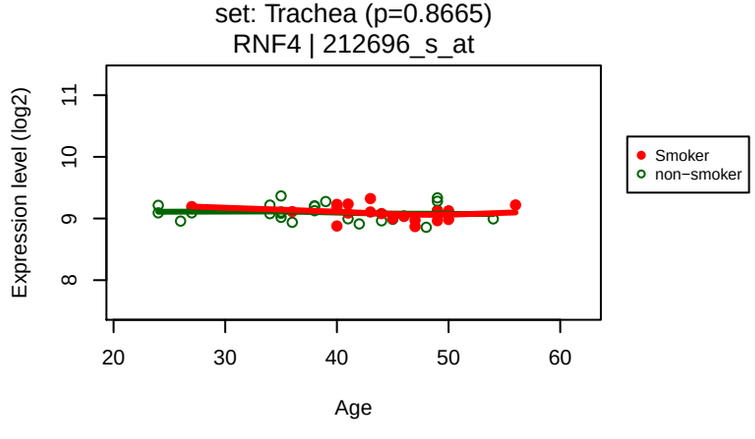

Gene expression ratio in Males
gene: RNF4 | 212696_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0295
p=0.2154
p=0.5145
p=0.6259
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNF4 | 212696_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8725
p=0.8342
p=0.986
p=0.6632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

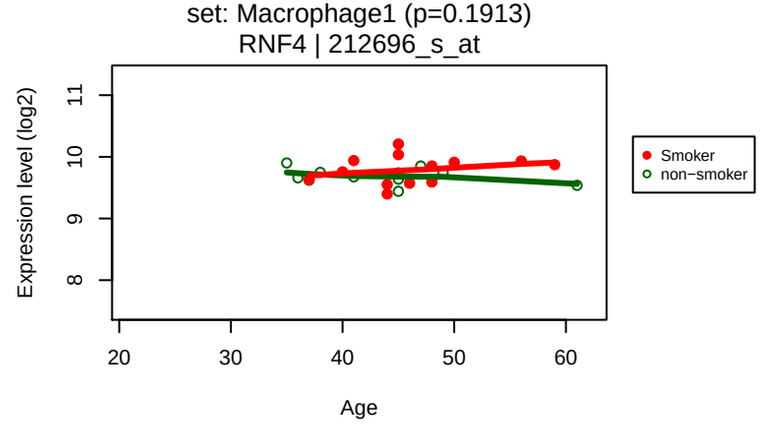

set: SmallAirway1 (p=0.0076)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker

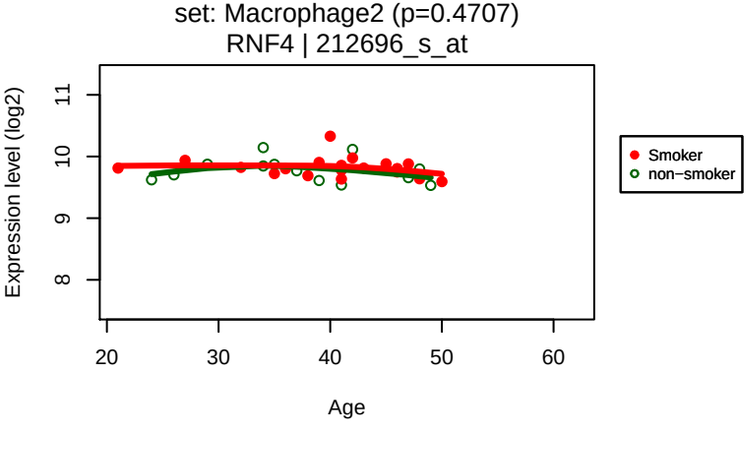

set: SmallAirway2 (p=0.0972)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker

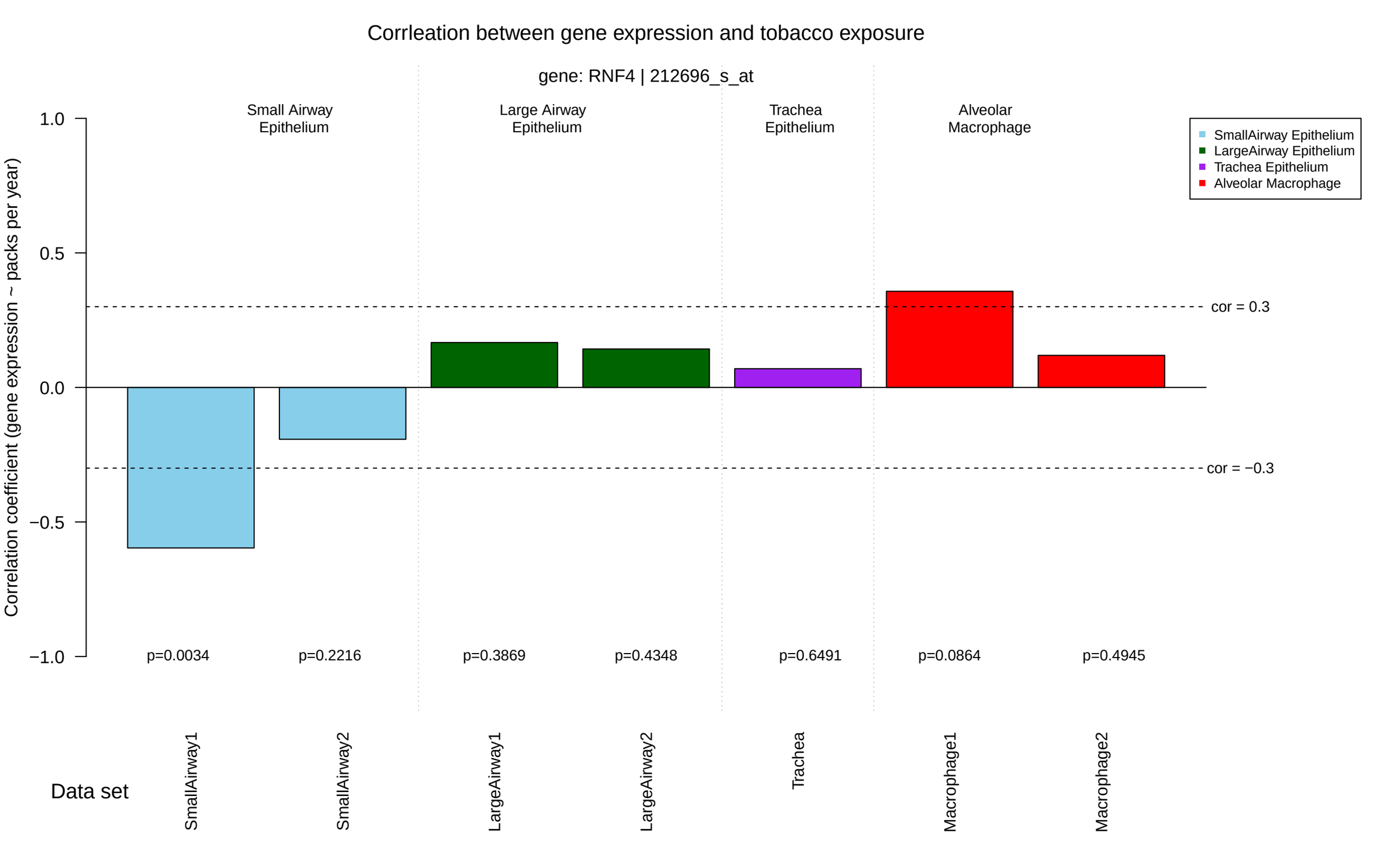

set: LargeAirway1 (p=0.8537)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4198)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8665)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1913)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4707)
RNF4 | 212696_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNF4 | 212696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0034
p=0.2216
p=0.3869
p=0.4348
p=0.6491
p=0.0864
p=0.4945
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2